

Structured summary: Power sensitivity of a hybrid open-label-plus-blinded- discontinuation N-of-1 design under carryover, serial correlation, and variance-component variation

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This document is a structured, section-by-section summary of the key points argued in the

35 companion manuscript, “Power sensitivity of a hybrid open-label-plus-blinded- discontinuation
36 N-of-1 design under carryover, serial correlation, and variance-component variation.” It
37 is provided as a supplementary reading aid and does not stand on its own; all notation,
38 derivations, simulation detail, and citations are in the main manuscript.

39

40 1 Introduction

41 **Aggregated N-of-1 designs offer power advantages over parallel RCTs, but the**
42 **conditions governing that advantage are incompletely characterized.**

- 43 • N-of-1 designs accumulate within-patient evidence across repeated crossover periods,
44 enabling meta-analytic population-level inference.
- 45 • Recent simulations report N-of-1 procedures matching or exceeding RCT power at
46 smaller sample sizes.
- 47 • The parameter axes that govern when the N-of-1 advantage holds or reverses are not
48 yet well specified.

49 **This paper reports a systematic sensitivity study of the Hendrickson hybrid N-**
50 **of-1 design for the main drug effect estimand.**

- 51 • The hybrid combines open-label titration, blinded discontinuation, and brief crossover
52 phases.
- 53 • Three comparators are evaluated: alternating A/B/A/B N-of-1, two-period crossover,
54 and parallel-group RCT.
- 55 • The target estimand is the main drug effect, distinct from the biomarker-treatment
56 interaction target examined elsewhere.

57 **Twelve sensitivity sweeps address the gap left by single-axis simulation compar-**
58 **isons in the published literature.**

- 59 • Published comparisons typically vary one or two parameters at a time, leaving the
60 primary power-ordering axes unidentified.
- 61 • The twelve sweeps span effect size, carryover half-life, decay-form misspecification, het-
62 erogeneity, within-patient variance, cycles, sample size, AR(1) correlation, period length,
63 biomarker architecture, a misspecification factorial, and null calibration.
- 64 • Each sweep axis is anchored to a specific precedent in the simulation-comparison litera-
65 ture.

66 **This paper extends the companion clinical comparison across twelve parameter**
67 **axes for the hybrid design family.**

- 68 • The companion paper provides a head-to-head comparison at the prazosin-calibrated
69 baseline.

- The present paper documents how that baseline comparison generalizes under systematic parameter perturbation.

1.1 Aims

The study follows the ADEMP framework with three aims: power characterization, design-ordering identification, and Type I error verification.

- Aim 1: characterize the hybrid’s power profile under systematic perturbation of published comparison axes.
- Aim 2: identify parameter regimes where the hybrid is preferred over the parallel RCT and where the ordering reverses.
- Aim 3: verify Type I error control with MCSE below 1%.

1.2 Pipeline

All sweeps use a baseline of $N = 35$ participants and 8 weekly timepoints, with sample size matched per design rather than per observation count.

- The pipeline is implemented in the `nof1power` R package, which re-implements the Hendrickson 2020 reference pipeline.
- The $N = 35$ minimum and per-design matching convention follow Blackston 2019 and Hendrickson 2020.
- Parameters are varied from this baseline one axis at a time across the twelve sweeps.

1.3 Dual data-generating architectures

Two data-generating architectures encode the biomarker interaction differently, producing substantively different power profiles under carryover.

- The covariance-moderation architecture (MVN) encodes the interaction in the covariance structure; on-drug correlation decays exponentially off-drug.
- The mean-moderation architecture shifts bio-response means additively at on-drug timepoints by the standardized biomarker.
- The two architectures are crossed in sweep S10; the remaining eleven sweeps use the covariance-moderation architecture as the default.

1.4 Response curves and carryover

Latent response factors follow a modified Gompertz curve; carryover is applied via a recursive decay from an extensible decay-family.

- The Gompertz curve passes through the origin and asymptotes to `maxr`, parameterized by displacement and rate.
- Off-drug carryover is applied recursively using one of four decay families: exponential, linear, Weibull, or power.

- The DGP decay family and the analysis-side assumption can differ; sweep S3 quantifies the cost of that mismatch.

1.5 Covariance structure and analysis

The analysis model is an `nlme::lme` with `corCAR1` residual correlation; the target estimand is the Dbc main drug effect coefficient.

- The covariance matrix couples within-factor AR(1), cross-factor correlations, and biomarker differential correlation; non-PD matrices are corrected with a per-cell flag.
- The fixed-effects formula is $Sx \sim bm + t + Dbc + bm:Dbc$ with a random intercept per participant.
- Singular fits are detected from the random-effects variance and flagged per replicate.

1.6 Reproducibility and execution

Per-replicate rather than pre-aggregated output is written, enabling Morris MCSE computation of all performance measures at render time.

- Each sweep seeds via `set.seed(42 + sweep_id)` and uses `furrr` multisession parallelism.
- RDS artifacts are written to `analysis/data/derived_data/sensitivity/` and orchestrated by `run-all.R`.
- Per-replicate storage allows power, bias, MSE, empirical SE, and Type I error to be computed with MCSE at render time.

2 Results

All twelve sweeps completed in approximately 32 minutes; per-cell metrics are computed at render time from per-replicate RDS artifacts.

- Wall-clock time was approximately 32 minutes on 8 cores with `furrr` multisession.
- Summary metrics are derived at render time from `analysis/data/derived_data/sensitivity/`.
- The target estimand throughout is the Dbc main drug effect coefficient.

The hybrid is the primary design; the alternating A/B/A/B variant is retained as a sensitivity comparator for sweeps that vary cycle count or period length.

- The hybrid's fixed eight-timepoint structure does not scale with cycle count or period length.
- The alternating variant is used in S6 and S9 where that scalability is required.

Across sweeps, the hybrid outperforms the parallel RCT on sample size, variance robustness, and serial-correlation tolerance; the RCT is more robust to long carryover.

- The hybrid reaches 80% power at smaller sample sizes, under higher within-patient variance, and under stronger AR(1) than the parallel RCT.

- The parallel RCT is more robust to carryover exceeding the within-subject period length and to decay-form misspecification.
- Type I error is near nominal for the hybrid and alternating variant; the two-period crossover shows robust inflation of 0.10 to 0.13.

2.1 S1. Effect-size sweep

The hybrid reaches 80% power at $\delta \approx -1$, a half-unit smaller effect size than the parallel RCT requires.

- Hybrid power rises from 0.066 at the null to saturation at 1.000 for $|\delta| \geq 2$.
- Parallel RCT power follows the same shape at a lower level, reaching 0.941 at $\delta = -2$.
- The hybrid's 80%-power effect-size threshold is approximately 0.5 units smaller than the RCT's.

Type I error is marginally above nominal for the hybrid and marginally below for the RCT; neither is grossly miscalibrated, but the hybrid's upward departure warrants attention.

- Hybrid Type I is 0.066 (95% CI [0.050, 0.082]); parallel RCT Type I is 0.034 (95% CI [0.022, 0.046]).
- Both confidence intervals cover 0.05 at the margin.
- The hybrid's modest upward departure merits caution when the study is powered near the 5% boundary.

2.2 S2. Carryover half-life sweep

The hybrid tolerates carryover up to 5 days before power erodes; the alternating variant degrades from 0.90 to 0.30 across the half-life range.

- Hybrid power is 1.000 through $t_{1/2} = 5$ days, declining to 0.736 at 14 days.
- The alternating N-of-1 is more sensitive, dropping from 0.896 to 0.304 across the same range.
- The parallel RCT is essentially immune, with power flat at 0.92 to 0.96 across the grid.

The hybrid's carryover robustness is attributable to its open-label titration phase; the RCT ordering reversal begins when $t_{1/2}$ exceeds the period length.

- The four-week open-label titration accumulates treatment signal that is not erased by the post-discontinuation off-drug period.
- The alternating design relies on repeated within-subject transitions and sheds signal proportional to the unwashed residual.
- The hybrid-versus-RCT power ordering crossover begins at approximately $t_{1/2} = 10$ days, consistent with Blackston 2019.

172 2.3 S3. Decay-form misspecification sweep

173 The decay-form sweep uses the alternating variant because the hybrid's multi-
174 phase `tsd` pattern produces non-PD matrices under the power decay form.

- 175 • Five DGP decay forms are tested while the analysis always assumes exponential decay.
- 176 • The hybrid is excluded because its multi-phase `tsd` structure interacts poorly with the
177 power form.
- 178 • The alternating variant provides a clean test of decay-form misspecification sensitivity.

179 Main-effect power ranges only 15 percentage points across decay forms, substan-
180 tially less sensitive to misspecification than the interaction estimand.

- 181 • Power ranges from 0.754 (Weibull $k = 0.8$) to 0.904 (linear) under the alternating N-of-1.
- 182 • The 15-point range is smaller than under the interaction target, indicating less sensitivity
183 to decay-form misspecification.
- 184 • The parallel RCT is flat at 0.93 to 0.95 across all forms, consistent with Gärtner 2023.

185 2.4 S4. Between-patient heterogeneity sweep

186 The hybrid is saturated across the τ range; the parallel RCT loses 12 percentage
187 points as between-patient heterogeneity doubles.

- 188 • Hybrid power is 1.000 at all six τ levels with stable mean $\beta \approx -0.94$.
- 189 • Parallel RCT power declines from 0.998 at $\tau = 0.5$ to 0.874 at $\tau = 2.0$.
- 190 • The RCT loses approximately 12 percentage points as τ doubles from 1.0 to 2.0.

191 The Senn 2018 variance-decomposition prediction is reproduced: τ^2 enters the
192 RCT estimator but is absorbed by the within-subject contrast in N-of-1 designs.

- 193 • Between-patient heterogeneity contributes to $\text{Var}(\hat{\Delta})$ in the parallel RCT but not in the
194 within-subject N-of-1 contrast.
- 195 • The hybrid's saturation reflects the prazosin-calibrated baseline where effect size is large
196 relative to within-patient variance.
- 197 • At smaller effect sizes or larger σ , the N-of-1 advantage would widen rather than flatten
198 as τ increases.

199 2.5 S5. Within-patient variance sweep

200 The hybrid is essentially saturated across the within-patient variance range; the
201 parallel RCT loses approximately 18 percentage points.

- 202 • Hybrid power declines from 1.000 to 0.986 across $\sigma \in \{0.5, \dots, 1.5\}$, a one-percentage-
203 point loss.
- 204 • Parallel RCT power declines from 0.996 to 0.814 across the same range, an 18-point
205 loss.
- 206 • The ordering is consistent with the N-of-1 design summing k paired contrasts per patient.

The hybrid's quasi-saturation is specific to the prazosin-calibrated regime; at smaller effect sizes the σ dependence would be steeper.

- The eight-timepoint structure plus open-label titration provides sufficient signal accumulation to absorb substantial within-patient noise.
- The Senn 2018 prediction is reproduced: N-of-1 absorbs within-patient variance via k paired contrasts.
- Saturation should not be extrapolated beyond the prazosin-calibrated effect-size regime.

2.6 S6. Cycles-per-patient sweep

Alternating N-of-1 power rises concavely with k , from 0.294 at $k = 2$ to 0.980 at $k = 16$, as predicted by the Senn 2018 decomposition.

- The alternating variant is used because the hybrid's fixed structure does not scale with k .
- Power rises from 0.294 at $k = 2$ to 0.804 at $k = 8$ (baseline) to 0.980 at $k = 16$.
- The concave shape is consistent with the Senn 2018 variance-decomposition prediction.

For the main-effect target, increasing k slightly favors the RCT; the hybrid's advantage over the alternating variant comes from the open-label phase, not cycle count.

- At $k = 2$ the alternating N-of-1 and the RCT are in a dead heat; at $k \geq 6$ the RCT surpasses the alternating N-of-1 by up to 14 points.
- Each additional timepoint contributes a between-subject observation for the RCT but averages into already-accumulated paired contrasts for the alternating N-of-1.
- The hybrid's primary power advantage over the alternating design arises from the open-label titration phase, not from the number of cycles.

2.7 S7. Patient-count sweep

The hybrid saturates at high power by $N = 16$; the parallel RCT reaches the same threshold between $N = 24$ and $N = 35$.

- Hybrid power is high at modest sample sizes and saturates rapidly with N .
- The parallel RCT follows a similar but offset trajectory, reaching 80% power between $N = 24$ and $N = 35$.
- The grid anchors the project-standard $N = 35$ minimum for cross-paper comparison.

The hybrid achieves an equivalent power target at approximately one-third to one-half the parallel-RCT sample size, consistent with Blackston 2019 and Carrozzo 2025.

- The hybrid reaches 80% power at $N \leq 16$; the parallel RCT reaches it between $N = 24$ and $N = 35$.

- The sample-size ratio of one-third to one-half matches the Blackston 2019 N-of-1 efficiency advantage.
- This is consistent with Carrozzo 2025, in which the meta-analytic N-of-1 required fewer participants across most heterogeneity and carryover settings.

2.8 S8. AR(1) serial-correlation sweep

The hybrid is insensitive to AR(1) correlation; the parallel RCT loses 24 percentage points of power as ρ rises from 0 to 0.9.

- Hybrid power is 1.000 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$.
- Parallel RCT power declines from 0.984 to 0.746, and mean $\hat{\beta}$ shrinks from -1.01 to -0.57 .
- The RCT loss is monotonic with ρ across the examined range.

The AR(1) asymmetry arises because the corCAR1 structure absorbs correlation within subjects but cannot manufacture the missing within-subject contrasts for the parallel RCT.

- Strong AR(1) produces coherent within-subject trajectories for the hybrid, absorbed by corCAR1 with negligible power loss.
- For the RCT, strong AR(1) inflates between-subject mean contrast variance via effective-sample-size shrinkage.
- This is consistent with Wang and Schork 2019: N-of-1 designs are insensitive to serial correlation, while parallel-group designs are not.

2.9 S9. Period-length sweep

Main-effect power increases monotonically with period length, from 0.40 at 3 days to 1.00 at 14 days; carryover half-life has only a modest effect at fixed period length.

- Power rises from 0.40 to 0.44 at 3-day periods to 0.997 to 1.00 at 14-day periods.
- At fixed period length, varying carryover half-life across 0.5 to 7 days produces only modest power differences.
- The sweep uses the alternating variant, as the hybrid's fixed structure does not scale with period length.

For the main-effect target, longer periods are preferred, reversing the interaction-target recommendation; carryover should not exceed approximately 50% of the period length.

- Longer periods accumulate more Gompertz drug-response signal per timepoint.
- The practical guideline is the reverse of the interaction-target recommendation: longer periods benefit main-effect detection.
- Wang and Schork 2019 note this tradeoff explicitly: frequent alternation benefits correlation-modeled analyses but not absolute-effect inference.

2.10 S10. Biomarker interaction strength crossed with DGP architecture

Main-effect power is invariant to c_{bm} and DGP architecture, confirming that the main drug effect is determined by Gompertz accumulation, not biomarker coupling.

- Hybrid power is 1.000 across all 16 c_{bm} -by-architecture-by-half-life combinations.
- Alternating N-of-1 power varies from 0.682 to 0.916; parallel RCT power varies from 0.914 to 0.958.
- The flat main-effect power across c_{bm} values provides a robustness check on the DGP implementation.

The flat c_{bm} power curves are a DGP implementation check: when the main effect is the target, biomarker coupling should not affect power, and it does not.

- The main drug effect is determined by Gompertz max[br] and on-drug accumulation, not by biomarker coupling.
- Approximately flat power across c_{bm} for all three designs confirms the DGP correctly separates the two signal sources.
- The reader should not conflate this main-effect invariance with the biomarker-interaction target, where c_{bm} is the primary power driver.

2.11 S11. Misspecification factorial

The parallel RCT shows statistically robust Type I inflation of 0.112 to 0.126 under high DGP AR(1); the hybrid is controlled near nominal across all four cells.

- Hybrid null Type I is 0.036 to 0.056, within 2 MCSE of nominal across all four DGP combinations.
- Parallel RCT Type I is 0.018 to 0.024 (conservative) when no AR(1) is present, and 0.112 to 0.126 (inflated) under $\rho = 0.7$.
- The inflation is statistically robust at MCSE 0.014 to 0.015.

The RCT Type I inflation arises because the between-subject variance estimator understates the true standard error when DGP serial correlation is strong.

- Under strong DGP AR(1), corCAR1 cannot manufacture the missing within-subject contrasts for a between-subject comparison.
- The hybrid is immune because its primary treatment contrast is within-subject.
- The mechanism is analogous to the Blackston 2019 carryover Type I inflation finding, though the source here is AR(1) rather than carryover.

2.12 S12. Null-effect calibration

Hybrid and alternating N-of-1 Type I error is near nominal; the two-period crossover shows robust inflation of 0.100 to 0.130 attributable to underidenti-

315 **fied corCAR1.**

- 316 • Hybrid Type I: [0.056, 0.062]; alternating N-of-1: [0.057, 0.064]; both within 2 MCSE
317 of nominal.
- 318 • Parallel RCT: [0.029, 0.060], three cells conservative and one nominal.
- 319 • Two-period crossover: [0.100, 0.130], statistically robust inflation at MCSE 0.006 to
320 0.007.

321 **The two-period crossover should not be analyzed with corCAR1; at two within-**
322 **subject timepoints the parameter is underidentified and the rejection rate inflates**
323 **by a practically meaningful amount.**

- 324 • Two within-subject timepoints cannot identify the corCAR1 parameter reliably.
- 325 • The inflation of 0.100 to 0.130 is statistically robust and practically meaningful.
- 326 • Alternative analyses (paired t -test on change scores or lme with unstructured residual)
327 are recommended for this design.

328 **2.13 Synthesis**

329 **Five substantive findings organize the twelve sweeps: sample-size advantage, vari-**
330 **ance robustness, carryover design-dependence, AR(1) asymmetry, and Type I**
331 **error calibration.**

- 332 • The hybrid reaches 80% power at $N \leq 16$; the parallel RCT requires $N = 24$ to 35, a
333 one-third to one-half sample-size ratio.
- 334 • The hybrid is saturated across σ and τ sweeps; the RCT degrades by 18 and 12 percent-
335 age points respectively.
- 336 • Carryover sensitivity is design-specific: the hybrid tolerates $t_{1/2} \leq 5$ days; the alternat-
337 ing variant drops from 0.90 to 0.30 across the range; the RCT is immune.

338 **2.13.1 Hendrickson et al. (2020): predictive biomarker validation**

339 **Hendrickson 2020 showed hybrid power collapsing from 0.95 to 0.18 at the**
340 **biomarker-interaction target under a 0.7-day carryover half-life.**

- 341 • At $N = 35$, $c_{bm} = 0.6$, no carryover, biomarker-interaction power was 0.95 (hybrid),
342 0.96 (crossover, CO), 0.94 (open-label plus blinded discontinuation, OL+BDC), and
343 0.25 (open-label, OL).
- 344 • A half-life of only 0.7 days collapsed hybrid power to 0.18, demonstrating extreme
345 carryover sensitivity at the interaction estimand.
- 346 • This contrasts sharply with the present study's main-effect findings, where hybrid power
347 is 1.000 through $t_{1/2} = 5$ days.

348 **The present study extends Hendrickson via Morris MCSE reporting, decay-form**
349 **misspecification (S3), and main-effect carryover and period-length quantification**
350 **(S2, S6, S9).**

- Design ranking at the biomarker-interaction target reproduces Hendrickson by construction.
- The key substantive divergence is estimand-specific: main-effect hybrid power is saturated through 5-day carryover, while interaction-target power collapses at under 1 day.
- Both findings are correct; they describe different estimands and must not be conflated.

2.13.2 Wang and Schork (2019): AR(1), heteroscedasticity, and period stacking
Wang and Schork 2019 show roughly linear power erosion with ρ for single-period designs; period stacking substantially mitigates this loss.

- For $1p \times 2i$, power erodes from 0.851 at $\rho = 0$ to 0.126 at $\rho = 0.75$.
- For $40p \times 2i$, the same ρ range produces only a 0.17-point power loss, confirming period stacking as a mitigation.

S8 is the aggregated-level analogue of Wang and Schork: the hybrid absorbs AR(1) via corCAR1; the parallel-RCT power loss directionally matches Wang and Schork's between-period profile.

- Hybrid power is saturated at 1.00 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ at the aggregated level.
- Parallel RCT power drops from 0.98 to 0.75, directionally consistent with Wang and Schork's between-period power loss.
- The S11 Type I inflation of 0.11 to 0.13 is consistent with the Wang and Schork mechanism, though magnitudes depend on small-sample instability of corCAR1 at $N = 35$.

2.14 Limitations

Four limitations apply: hybrid saturation at the prazosin baseline, alternating-variant substitution in three sweeps, analysis-only corCAR1 in S8, and small-sample instability in the S11 RCT Type I estimate.

- Saturation in S2, S4, S5, S8, S10, and S11 prevents quantifying the hybrid's relative advantage in sub-ceiling regimes; sweeps at smaller effect sizes are the priority extension.
- S3, S6, and S9 use the alternating variant because the hybrid cannot scale with decay form, cycle count, or period length; results apply to the alternating family, not directly to the hybrid.
- The S11 RCT Type I ranges (0.112 to 0.126 vs. 0.029 to 0.060 in S12) are not contradictory; they reflect different DGP regimes.

2.15 N-of-1 design variants not evaluated

Several N-of-1 design variants are not evaluated; these fall into designs expected to be weaker than those reported here and designs expected to be stronger.

- Unevaluated designs include single-subject, OL-only, OL+BDC-only, full multi-cycle crossover, complete crossover, sequential-monitoring, and randomized-block variants.

- Single-subject and reduced variants are expected to produce power at or below the reported numbers; results are an upper bound.
- Full multi-cycle, complete-design, and sequential-monitoring variants are expected to exceed the reported numbers; results are a lower bound.

The present results bound the unevaluated variants: weaker designs are upper-bounded and stronger designs are lower-bounded by the reported numbers.

- Simpler variants (single-subject, OL-only, OL+BDC-only, two-cycle crossover) are expected at or below the reported power.
- More powerful variants (multi-cycle, complete-design, sequential-monitoring, randomized-block) are expected at or above the reported power.
- Extending the present framework to cover these variants is a tractable implementation task rather than a conceptual redesign.

2.16 Prototype Monte Carlo confirmation

A 500-replicate prototype ran in 191 seconds with 100% convergence; the MCSE on a power estimate near 0.3 is approximately 0.020.

- The 15-cell $c_{br} \times N$ prototype grid was completed in 191 seconds on 9 cores.
- Convergence was 100% across all 7,500 fits.
- Per-replicate output and Morris ADEMP summary are at `analysis/data/quick-sim/`.

The apparent non-monotonic power dependence on c_{br} from the 100-replicate prototype does not survive at 500 replicates; the signal is broadly flat.

- At $N = 25$, the five-level swing is 0.024 (one MCSE), consistent with sampling noise around a constant.
- At $N = 35$ and $N = 50$, the peak-to-trough swings of approximately three MCSE are suggestive but not decisive, and the peaks do not coincide on c_{br} .
- The 100-replicate dip at $c_{br} = 0.15$ has disappeared, confirming it was sampling noise.

The dominant axis is sample size, not c_{br} ; no prototype cell attains 80% power, and the grid does not bound the 80%-power region.

- Mean power rises from 0.18 at $N = 25$ to 0.31 at $N = 50$; the c_{br} effect is a possible shallow optimum near $[0.30, 0.45]$.
- No (c_{br}, N) cell in the prototype reaches 80% power; a production run requires larger N , larger $br_{\max}$, or both.
- The prototype grid does not bound the 80%-power region, which is the central deliverable for trial planning.

3 Conclusions

Three conclusions: the hybrid is a competent main-effect default, design ordering is estimand-dependent, and two-period crossovers should not be analyzed with `corCAR1`.

- The hybrid absorbs variance and serial correlation, tolerates carryover up to five days, and achieves 80% power at smaller sample sizes than the parallel RCT.
- Design ordering is substantially estimand-dependent: the hybrid's advantage is larger for biomarker interactions than for main effects.
- The two-period crossover's robust Type I inflation of 0.10 to 0.13 warrants a change to standard analytic practice when `corCAR1` is fitted at two within-subject timepoints.

The hybrid N-of-1 is supported as a default candidate for trial planning in chronic neurobehavioral conditions where moderate treatment heterogeneity and short-to-moderate carryover are expected.

- The companion clinical paper provides the baseline head-to-head comparison; this paper documents its generalization across twelve parameter axes.
- Robustness across most axes, with the identified exceptions for long carryover and two-period crossover analysis, supports the hybrid as a default candidate.

4 Data availability statement

Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are versioned in the `pmsimstats-ng` repository; exact paths are in the Reproducibility section.

- Monte Carlo outputs are under `analysis/data/`.
- Simulation driver scripts are under `analysis/scripts/`.

5 Reproducibility

The manuscript was rendered with R 4.4.x inside the `zzcollab` Docker container; principal packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.

- Image hash is in `Dockerfile`; package set is in `renv.lock`.
- `nlme` provides `corCAR1` residual correlation; `furrr`/`purrr` provide the tidyverse collection; `data.table` provides the original-extended collection.

Two-level seeding ensures reproducibility: driver-level seeds govern parameter grids; replicate-level seeds derive from a master seed via the replicate index.

- Driver level: `set.seed(42 + sweep_id)` before parameter grid construction.
- Replicate level: `set.seed(20260507)` at the loop top, with `parallel::mc.set.seed` inside `mclapply`.

453 • The two seeding stages operate at distinct pipeline points and do not collide.

454 **All file paths for reproducing this manuscript are fixed: per-replicate RDS,**
455 **ADEMP summary, driver scripts, pre-registered plan, and manuscript source.**

456 • Per-replicate outputs: analysis/data/quick-sim/05-design-replicates.rds;
457 ADEMP summary: 05-design-summary.txt.

458 • Driver scripts: analysis/scripts/nof1-design-sensitivity/ and analysis/scripts/quick-sim/05-

459 • Pre-registered ADEMP plan: 00-ademp-pre-registration.md; manuscript source:
460 analysis/report/05-nof1-design-sensitivity/.